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Shape stabilized phase change materials based on different support structures for thermal energy storage applications–A review

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ABSTRACT

Thermal energy storage systems play a crucial role in energy conservation and balancing energy demand/supply. Recent thermal storage techniques and novel strategies have expanded their usage in various applications. However, leakage during phase change and poor thermal conductivity limits using phase change materials (PCM) as a potential thermal storage medium. Shape-stabilized phase change materials (SSPCM) can effectively enhance heat transfer and prevent leakage. Besides, it provides flexible structures, good mechanical strength, and stability. Furthermore, loading a maximum quantity of PCM in the support structure enables improved efficiency of SSPCMs and enhances heat transportation. In this review work, SSPCMs and different types of support structures used to prepare SSPCM are discussed and presented with their advantages and disadvantages. It is also aimed to provide comprehensive information on microencapsulation techniques, metallic, carbon-based, and polymeric support employed in SSPCM preparation. This review also sheds some light on the applications of SSPCM, more specifically, thermal management and storage. Finally, the future scope of research on SSPCM is briefly discussed. It is believed that the information presented in this review will help the readers to understand SSPCM and different support structures for SSPCM preparation, along with various application techniques.

1. Introduction

Energy consumption due to human activities shows a continuous increment owing to various upgraded lifestyles and technologies. In order to reduce energy consumption and utilize the available energy effectively, various energy conservation measures are being adopted in all sectors. Energy storage is a potential solution that can address energy consumption issues by balancing energy supply and demand. The adverse effect on the environment caused by fossil fuels instigated the renewable energy practice. But the low energy density and intermittent nature of renewable energy resources limit their application for various applications. By implementing an efficient energy storage technology, renewable energy utilization can be largely expanded.

The various energy storage methods are mechanical energy storage, electrical energy storage, thermal energy storage, etc. It is well known that thermal energy is one of the most significant end-use energy forms. Therefore, storing thermal energy will provide significant support to energy conservation. Thermal energy storage can be classified based on types of thermal energy as thermochemical storage, sensible storage, and latent heat storage [1]. Thermal energy is stored through reversible chemical reactions involving high energy in thermochemical storage. The reaction potential of the chemical reaction is the measure of stored thermal energy [2]. In sensible storage, the thermal energy is stored by altering the temperature of the storage medium. Therefore, the mass of the energy storage medium, specific heat capacity, and temperature change determines the amount of energy storage. Lastly, in latent heat storage, the thermal energy is stored during the phase change process of the energy storage medium, which is termed phase change material (PCM). Latent heat storage with PCMs is widely used due to advantages such as high energy storage capacity, constant temperature heat storage and retrieval, and comparatively less space occupancy [3]. There are three types of PCMs based on their phase transformation: solid to solid, solid to liquid, and liquid to gas. Solid to liquid PCMs are widely used among the different types due to their large energy storage capacity and low volume change during energy storage and retrieval [4,5].

In recent times, PCMs have been employed in various applications such as solar thermal appliances [6,7], thermal comfort [8–10], cold chain supply [11], perishable food storage [12,13], freezers [14], and transportation to overcome the problem of the gap between energy

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Abbreviations

CNT	Carbon nanotube
CPCM	Composite phase change material
DSC	Differential scanning calorimetry
EG	Expanded graphite
EAFD	Electric arc furnace dust
MEPCM	Microencapsulated phase change material
MPS	3-(trimethoxysilyl) propyl methacrylate
MWCNT	Multiwalled carbon nanotube
PCM	Phase change material
PMMA	Polymethyl methacrylate
PEG	Polyethylene glycol
SSPCM	Shape stabilized phase change material
SEM	Scanning electron microscopy
SWCNT	Single walled carbon nanotube
TES	Thermal energy storage
TGA	Thermogravimetric analysis
VTMS	Vinyl trimethoxy silane

demand and supply. However, there are practical problems when applying PCM in conventional methods. For example, there exist difficulties such as leaking, volume change during phase transformation, and low heat transfer due to poor thermal conductivity. To avoid these practical problems in PCM application, researchers are working toward novel strategies to make an efficient thermal energy storage system. Generally, solid to liquid PCMs are encapsulated in containers with different geometry. Leaking is one of the major issues when the PCM undergoes a phase transformation from solid to liquid. Shape stabilization is a promising solution to address the leakage problem while using the PCM. Shape stabilized phase change materials (SSPCMs) are energy storage materials stored in a support structure that can be used for various applications. The support structure will hold the PCM during the phase transformation and prevent leaking as the melted PCM is confined inside the structure. The materials used for the support structure will also increase thermal conductivity, which helps in efficient heat storage and retrieval. Generally, organic PCMs possess low thermal conductivity, and it creates a problem with heat transportation while using them in the application. Shape stabilization will help to address this issue by improving the heat transfer rate. However, the reduction in energy storage density is an adverse effect of shape stabilization. Developing an optimized SSPCM makes the thermal energy storage system more stable. Novel, highly efficient SSPCMs expanded the applications of PCMs in different areas due to the flexibility in operation.

Solar thermal energy harvesting and thermal energy management are the two major areas where SSPCM plays a significant role. Efficient thermal energy storage techniques can improve the renewable energy harvesting potential. Sadeghian et al. [15] presented a study on the effect of ribs on the cooling of photovoltaic modules. They confirmed that lowering the PCM melting point increases the electrical efficiency and output power of the photovoltaic panel. Yang et al. [16] fabricated a flexible SSPCM for thermal management with passive radiative cooling capacity. It can be used in electronic devices and buildings with excellent mechanical properties. Yeh et al. [17] experimentally and numerically analyzed the performance of SSPCM integrated with the solar thermal collector. Besides, Mehrali et al. [18] developed salt hydrate-based SSPCM for simultaneous solar thermal energy harvesting and storage with high thermal conductivity and photothermal efficiency.

This review is based on the different SSPCM support structures for thermal energy storage (TES) applications. This work is structured in such a way to provide comprehensive insight into the various types of support structures, such as metallic, carbon-based, polymer matrix, and microencapsulation techniques used for SSPCM and the role of SSPCM in different TES applications. It also summarizes the issues like thermal conductivity, degree of supercooling, stability of SSPCM, and application strategies.

2. Support structures for SSPCM

The SSPCM consists of two parts. One is the PCM which stores the energy, and the other is the support structure that holds the PCM. The support materials used for shape stabilization are of different types: microencapsulation, metallic support (metal form/matrix), carbon-based (nanostructure), and polymer matrix, as shown in Fig. 1. Several studies are available on the preparation of SSPCMs with different support structures and are focused on developing SSPCM with high energy density and high thermal conductivity. In addition, researches focus on increasing the thermal conductivity and developing multi-functional SSPCM for the smart TES system. In this section, as classified in Fig. 1, the support structures of SSPCM that have been recently studied are introduced and summarized.

2.1. Microencapsulation

Microencapsulation is a technique in which the PCMs are covered using shell material of a smaller size. Microencapsulated phase change materials (MEPCM) are prepared by three methods, namely physical, chemical and physico-chemical methods. The different techniques of PCM microencapsulation are presented in Fig. 2.

Since the shell materials decide the performance of MEPCM, selecting a suitable shell material is essential for specific applications. The shell material has different types: organic, inorganic, and hybrid. There are several desirable properties for selecting a shell material, as below.

- Good mechanical strength
- Resistant to corrosion
- Good thermal stability according to the operating temperature
- Good thermal conductivity
- Good chemical/structural stability
- Should not react with the PCM core
- Easy to handle

2.1.1. Organic shell

The organic shell materials used for microencapsulation of PCMs are generally organic resins. Due to their flexible nature, organic shells provide structural flexibility during the phase change process but are not thermally and chemically stable. Some organic shell materials contain formaldehyde, which affects health and negatively impacts the environment. In addition, some organic polymer shells are flammable and brittle, restricting their encapsulation usage. Polymethyl methacrylate (PMMA) is a synthetic resin and has properties suitable for encapsulation material. Sari et al. reported on microencapsulated n-octacosane [19] and n-heptadecane [20] using PMMA, and the result showed good thermal and chemical stability. Alkan et al. prepared microcapsules of n-eicosane [21] and docosane [22] through emulsion polymerization using PMMA as shell material. The potential of the prepared microcapsules was studied for energy storage applications. Lashgari et al. [23] prepared n-hexadecane microcapsules using PMMA and poly butyl acrylate-co-methyl methacrylate shell material through suspension polymerization. The flexibility of the shell material and thermal performance were investigated. Wang et al. [24] fabricated a MEPCM with n-octadecane and PMMA with a thermochromic function. They used the suspension polymerization technique to prepare bifunctional microcapsules, which can be applied to various energy storage applications. Sari et al. [25] developed thermally stable microcapsules of myristic acid-palmitic acid eutectic PCM using a PMMA shell and tested up to 5000 heating and cooling cycles. Zhang et al. [26] prepared stearic

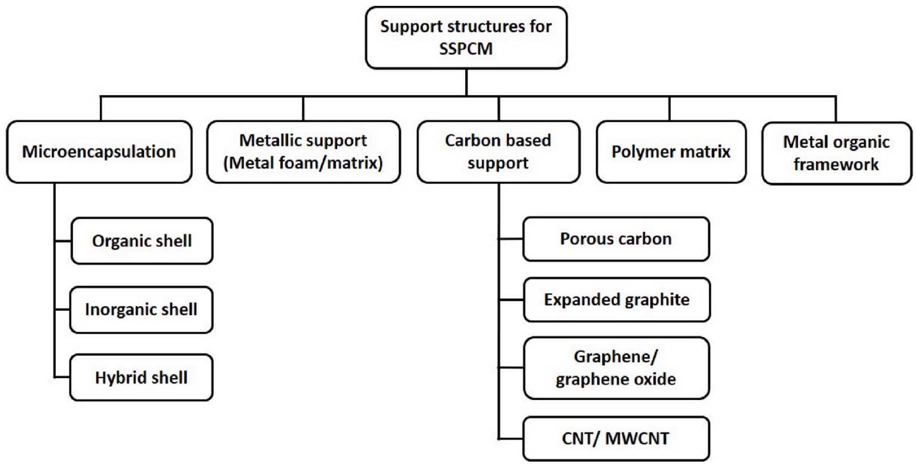


Fig. 1. Classification of SSPCM based on the support structure.

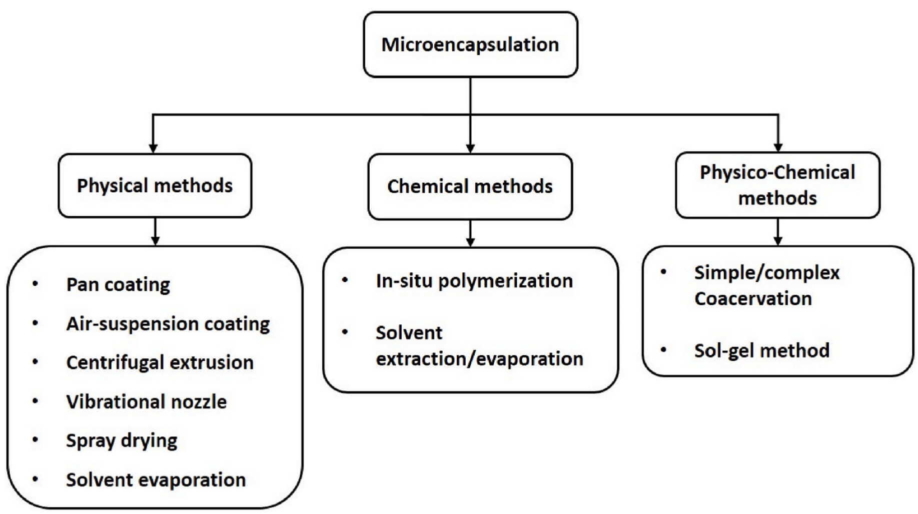


Fig. 2. Different techniques of PCM microencapsulation.

acid/PMMA microcapsules with excellent thermal reliability and chemical stability using iron (III) chloride as a photoinitiator through emulsion polymerization. Through suspension polymerization, paraffin was microencapsulated using PMMA shell material by Al-Shannaq et al. [27], resulting in a regular shape and smooth surface for a 2:1 core/shell mass ratio. The SEM micrographs of PCM microcapsules with different core/shell mass ratio is shown in Fig. 3.

Capric acid-palmitic acid eutectic PCM microcapsules were prepared by Xing et al. [28] with a shell core ratio of 1:2 through the solvent evaporation method and studied using different characterization techniques for TES applications. In addition, Polyvinyl chloride was used as shell material. Sanchez-Silva et al. [29] investigated the physical and thermal properties of microencapsulated paraffin PCM through suspension co-polymerization. Styrene-methylmethacrylate was used as shell material. N-heptadecane was microencapsulated using starch shell material by Irani et al. [30] for textile and building surface coating applications. Various properties were investigated for different core-shell weight ratios, mixing, and reaction times during the preparation. Chen et al. [31] fabricated and studied the thermal properties of octadecylamine-grafted graphene oxide PCM microcapsules using melamine-formaldehyde shell material. The results showed that the prepared microencapsulated PCMs possess excellent thermal properties in storing and retrieving thermal energy.

2.1.2. Inorganic shell

The commonly used inorganic shell materials are silica, titania, zinc oxide, and calcium carbonate [32]. The inorganic shell materials are thermally stable and possess good thermal conductivity compared to organic shell materials. Tahan et al. [33] prepared stearic acid nanocapsules using titania shell through the sol-gel method and estimated the melting temperature range and latent heat capacity range of 54–56 °C and 50–109 Jg^{-1}, respectively. The prepared microcapsules possess good thermal conductivity and stability. A novel bifunctional MEPCM of n-eicosane in titanium dioxide shell material was developed by Chai et al. [34]. The microscopy studies showed that the microcapsules are in a perfectly spherical shape with a size of 1.5–2 μm. The prepared microcapsules exhibited good photocatalytic effects, antimicrobial properties, and high thermal reliability of heat storage and retrieval. Zhao et al. [35] fabricated n-octadecane/TiO_2 microcapsules with an average size of 2–5 μm through a simple, environmentally friendly process and studied the properties using various characterization techniques. The microcapsules showed good thermal performance. Liu et al. [36] prepared a dual responsive MEPCM with n-eicosane and ZnO doped TiO_2 shell material through emulsion templated interfacial polycondensation, as shown in Fig. 4. The prepared microcapsules were studied by incorporated into gypsum boards, and they showed excellent thermal reliability, operation durability, and enhanced antibacterial properties. In their other work, the same authors fabricated microcapsules of

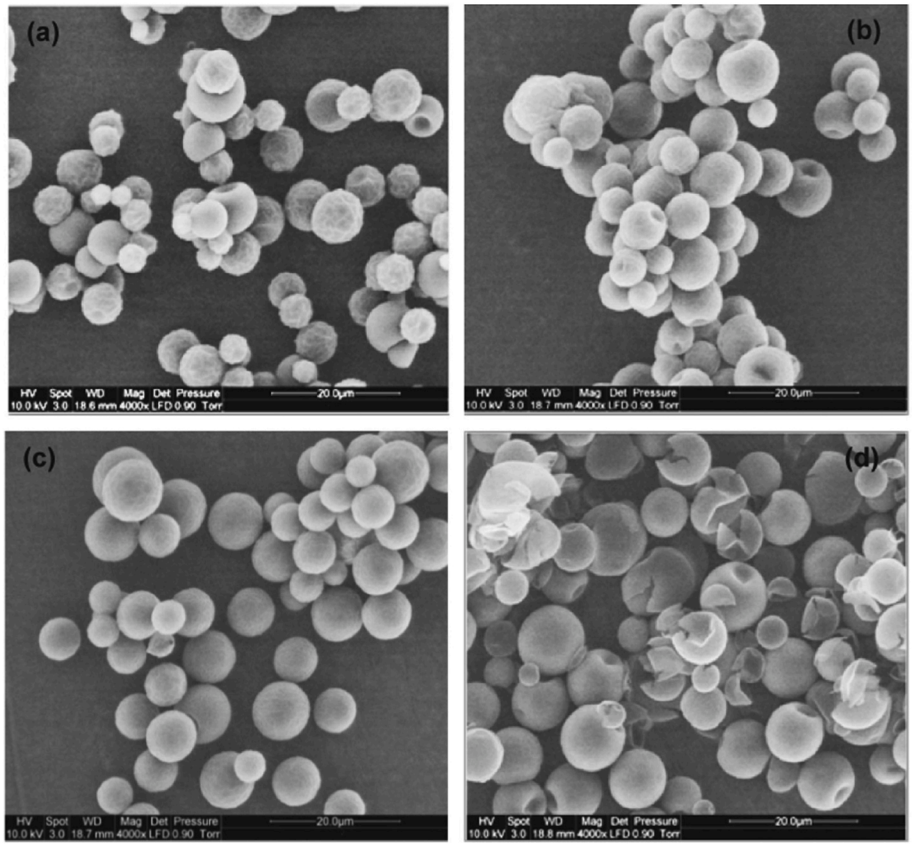


Fig. 3. SEM micrographs of PCM microcapsules with different core/shell mass ratio (a) 1:1, (b) 1.5:1, (c) 2:1 and (d) 3:1 [27].

n-eicosane/TiO_2 in three different morphologies, namely tubular, octahedral and spherical, using an emulsion templated interfacial polycondensation process [37]. Among the different microcapsules, the tubular structure shows better thermal response, and spherical microcapsules show high storage capacity due to the high core encapsulation ratio. The microstructure of different morphologies is shown in Fig. 5.

Using the sol-gel method, octadecane/silica microcapsules were synthesized by Tang et al. [38], and their various properties were investigated. The result showed that the MEPCM has a melting and freezing point of 28.32 °C and 26.22 °C and corresponding latent heat values of 227.66 $kJkg^{-1}$ and 226.26 $kJkg^{-1}$. It also possesses good thermal stability. Stearic acid was encapsulated using silica shell material through the sol-gel process by Yuan et al. [39]. The latent heat of the encapsulated PCM was 169.4 $Jg^{-1,}$ and the thermal stability of the PCM was improved by a silica shell. Sun et al. [40] developed polyethylene glycol (PEG)/silica microcapsules with well-defined core-shell structures using reverse emulsion-templated in-situ polycondensation. About 80% encapsulation ratio was achieved with a latent heat capacity of 130 $Jg^{-1,}$ and high heat storage efficiency was evidenced. By using interfacial polycondensation, Zhu et al. [41] prepared and studied the thermal properties of stearyl alcohol/SiO_2 microcapsules with different core-shell ratios. The MEPCMs made from 10 g of stearyl alcohol and 10 g of tetraethoxysilane exhibited good thermal stability after 100 thermal cycles and are suitable for TES application. The thermal conductivity at room temperature was estimated as 0.1412 $Wm^{-1}K^{-1}$. Ishak et al. [42] studied the effect of the core-shell ratio of lauric acid/SiO_2 microcapsules, and the encapsulation efficiency was found to be 93.48%, with good thermal stability and reliability. Microcapsules of capric acid-stearic acid eutectic PCM with silica shell was fabricated by Song et al. [43]. The melting and freezing temperatures were 21.4 °C and 22.2 °C, and the latent heat of melting and freezing were 91.48 Jg^{-1} and 90.52 $Jg^{-1,}$ respectively. The prepared microcapsules were thermally stable up to 1100 thermal cycles. In addition, Microcapsules of n-eicosane with antimicrobial properties were prepared by Zhang et al. [44] with a silver/silica double-layered shell. The DSC studies showed the presence of 67 wt% of n-eicosane and 33 wt% of shell material in the microcapsules. The electrical conductivity of the microcapsules was estimated as 130 Ωm. Wu et al. [45] synthesized mannitol/silica core-shell microcapsules through interfacial polymerization reaction with a phase change temperature range of 142.1–166.2 °C and latent heat of 147.4 Jg^{-1}. The microcapsules were stable, and no leakage happened during the phase change process.

2.1.3. Hybrid shell

To address the problems in organic shell materials, some inorganic additives are added to them, and several studies showed that superior thermal properties could be achieved. Ma et al. [46] prepared a hybrid shell microcapsule of oxalic acid/boric acid eutectic PCM using ethylcellulose, acrylonitrile butadiene styrene, and polydimethylsiloxane through an eco-friendly phase separation process. This method of microencapsulation is more suitable for water-soluble PCMs. Yang et al. [47] studied the organic PMMA shell material with inorganic silicon nitride for microencapsulating n-octadecane. With the addition of 10 wt % of silicon nitride, the thermal performance was enhanced, and mechanical properties were improved 4 times higher than that of PCM. Stearic acid was microencapsulated using PMMA hybrid shell material by Sahan et al. [48] through the emulsion polymerization technique. The developed microcapsules were thermally and chemically stable. Li et al. [49] prepared n-octadecane microcapsules with excellent thermal performance using PMMA/TiO2 hybrid shell material. TiO_2 particles improved thermal conductivity and stability. A composite titania-polyurea shell microcapsules of n-octadecane with 73 wt% of the core material was prepared by Zhao et al. [50] through interfacial polymerization. The MEPCM showed a better thermal response and

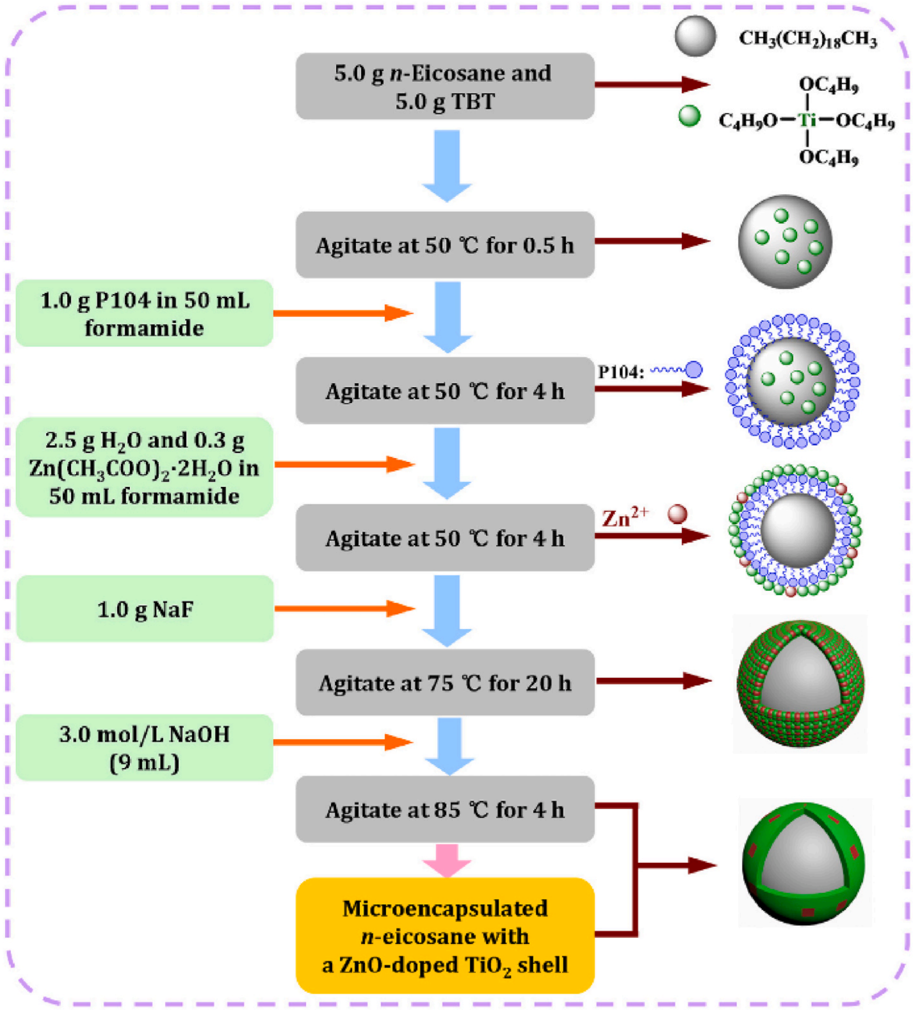


Fig. 4. Scheme of the synthetic route of microencapsulated n-eicosane with ZnO doped TiO_2 shell [36].

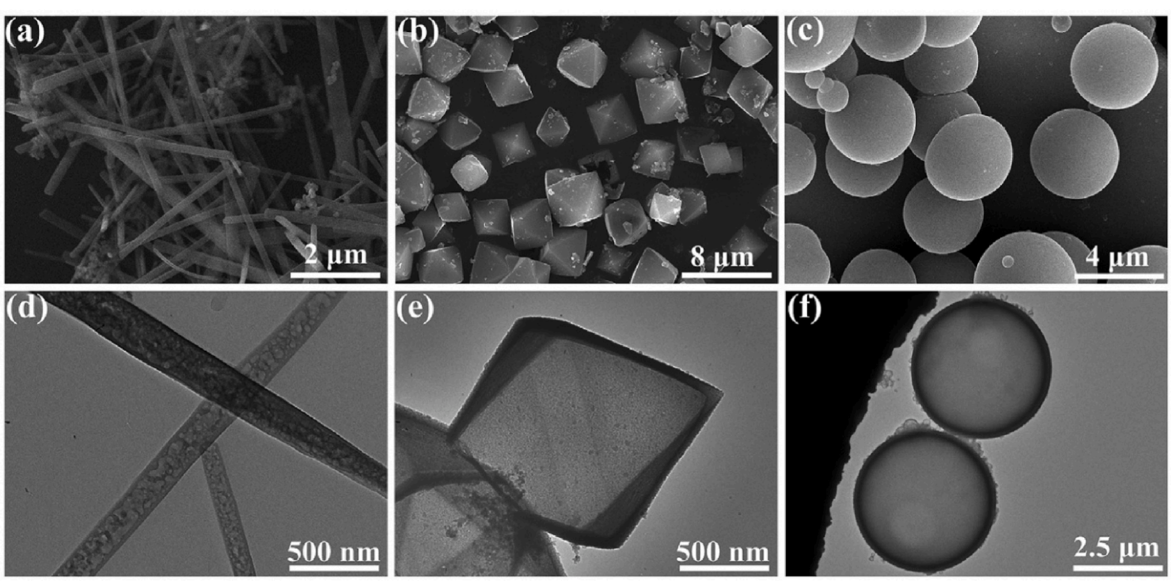


Fig. 5. SEM micrograph (a) Tubular, (b) Octahedral and (c) spherical microcapsules, TEM micrograph (d) Tubular, (e) Octahedral and (f) spherical microcapsules [37].

latent heat capacity. Jiang et al. [51] fabricated MEPCM of n-eicosane with TiO_2/Fe_3O_4 hybrid shell for thermoregulatory bio-applications. The microcapsules prepared through this method exhibited higher thermal stability and better reusability. A novel encapsulated PCM with SiO_2/TiO_2/Polydopamine composite shell material was prepared by Pornea and Kim [52] through the interlayer arrangement. The prepared material possesses photocatalytic activity, light scattering effect, and visible light absorption capacity. Gao et al. [53] fabricated paraffin wax microcapsules with styrene-divinylbenzene-silica (St-DVB-SiO_2) hybrid shell material. The microcapsules showed a melting temperature range of 44.36–55.03 °C and latent heat of 38.9–95.67 Jg^{-1}. Microcapsules of n-octadecane with PMMA/SiO_2 hybrid shell were developed by Wang et al. [54] through a low energy consumption photocurable polymerization process, as shown in Fig. 6. The encapsulation efficiency was 62.55%, with a melting latent heat of 122.31 Jg^{-1}. The prepared microcapsules exhibited excellent heat storage and release capacity, thermal reliability, and stability. The SEM micrographs of the microcapsules are also shown in Fig. 7.

Xia et al. [55] developed a MEPCM of n-octadecane with a melamine-formaldehyde/boron nitride hybrid shell through in-situ polymerization. The results showed excellent thermal stability and improved thermal conductivity. The latent heat of melting and freezing was 145.5 Jg^{-1} and 127.9 Jg^{-1}, respectively. The micro-PCM was stable on repeated thermal cycling and can be used for solar energy harvesting. Yin et al. [56] prepared MEPCM of 1-dodecanol with polymer-silica hybrid shell through Pickering emulsion. The latent heat of the prepared MEPCM was measured as 125 Jg^{-1} and showed good durability and stability. The utility efficiency of dodecanol was 94.2%. Through Pickering emulsion polymerization process, polymer-SiO_2/TiC hybrid shell micro-PCM of octadecane was prepared by Wang et al. [57]. Modified SiO_2 and TiC nanoparticles were used with MMA polymer for the preparation process, and the micro-PCM fabrication process is shown in Fig. 8. The results showed enhanced thermal stability and improved thermal conductivity. Hybrid shell micro-PCMs of n-octadecane with 3-(trimethoxysilyl) propyl methacrylate (MPS) and vinyltrimethoxysilane (VTMS) were developed by Li et al. [58]. At a weight ratio of 2:6 (MPS:VTMS), the micro-PCM possesses the highest latent heat of melting and freezing of 166.74 Jg^{-1} and 169.35 Jg^{-1}, respectively. The PCM content in the microcapsules was reduced by 7 wt% and 10.8 wt% after thermal treatment and thermal cycling.

Summarized microencapsulated SSPCMs from the literature are presented in Table 1. In order to improve encapsulation efficiency and energy storage capacity and improve the performance of PCM microcapsules as well as prevent the core from leaking and flowing out of the microcapsules, numerous research groups have looked into the synthesis methodologies, microstructures, and a variety of MPCM properties with various core and shell materials. As a result, various PCM-based microcapsules with a high TES potential and long-term durability have been developed through the study and development of the microencapsulation technology. By choosing the right wall materials and managing the synthetic conditions, several promising developments have been made for these microcapsules, providing them with a considerably wider variety of uses for TES. The literature has shown that polymeric shell materials are the interest of most researchers due to their effective protection and good stability with a high surface-to-volume ratio. Besides, organic PCMs are encapsulated more due to their less reactiveness with the shell materials and stability. The previous research on microencapsulation techniques showed that both organic and inorganic shell materials are extensively investigated for PCM-based TES applications. Besides, it can be seen that the organic-inorganic hybrid shell materials exhibit superior thermal properties to the individual organic and inorganic shell materials. This makes the hybrid shell materials attractive among researchers, and more intensive research is essential to explore their advantages and significant applications. The selection of specific preparation techniques for MPCMs is based on different parameters such as core and shell materials, size and thickness of the shell, properties of shell materials, etc. However, the chemical method of MPCM preparation is widely used due to less energy consumption and high yield. Among the different polymerization techniques, emulsion polymerization and interfacial polymerization were most used for the microencapsulation of PCMs. MPCMs with less core-shell ratio and less microencapsulation yield are limitations in the MPCM preparation process. New techniques must be further explored to achieve a high core-shell ratio with excellent thermal and physical properties.

2.2. Metallic support structures (metal foams/matrix)

Metal foams with high porosity, thermal conductivity, and strength can be used as a promising support structure for PCMs used for various applications. The addition of metal foams can significantly improve thermal conductivity. Different types of metal foams commonly used for SSPCM preparation are copper foams, aluminum foam, nickel foam, and stainless-steel foam. Xiao et al. [74] presented a novel copper-coated melamine foam-based PCM for the intelligent thermal management system. The PCM possesses a high thermal conversion efficiency of 85.6% with good stiffness, and its application to electronic device thermal management was presented. Li et al. [75] developed an SSPCM by impregnating erythritol in the surface roughened copper foam, and as a result, the thermal conductivity and supercooling were enhanced. The heat extraction efficiency and latent heat-releasing percentage were estimated as 94.2% and 85.8%, respectively. Composite PCM with highly porous copper foam and paraffin was prepared by Sheng et al. [76] with improved thermal conductivity of 3.29 $Wm^{-1}K^{-1}$. In their study, the copper foam was developed by simple solution combustion synthesis using copper nitrate and starch.

Metal foam/paraffin composite was prepared, and a detailed experimental study was performed by Yao et al. [77]. The result showed that the decrease in pore density results in enhanced thermal diffusion, fast-propelled phase interface, improved phase change synchronism, and temperature uniformity of metal foam/paraffin composite. Lafdi et al. [78] performed an experimental study to investigate the porosity and pore size of the aluminum foam during the melting of a composite

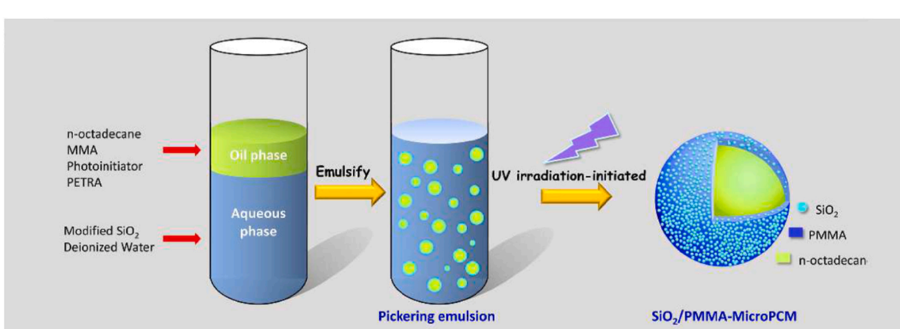


Fig. 6. Schematic illustration of pickering photocurable polymerization process [54].

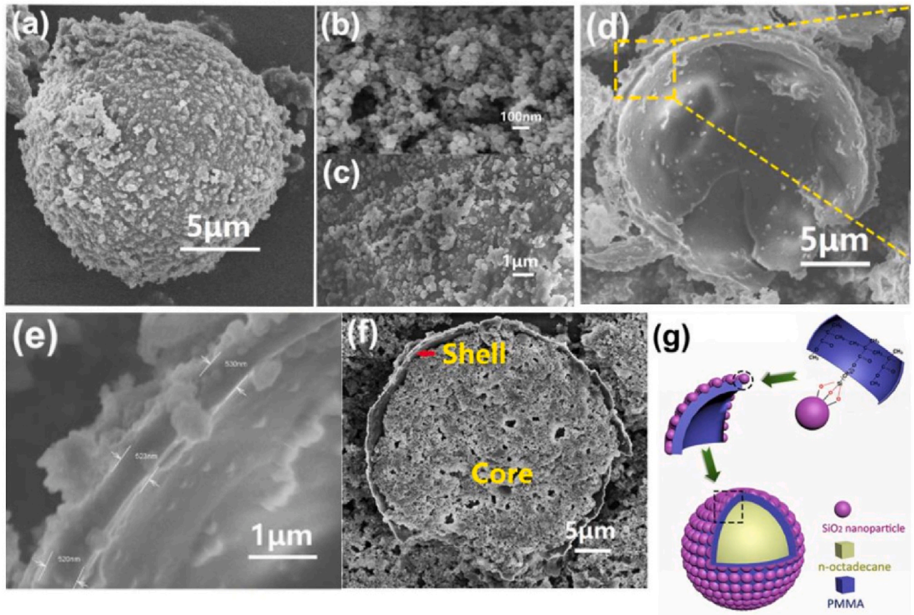


Fig. 7. SEM micrographs: (a) magnified MicroPCM, (b) SiO_2 nanoparticles, (c) Surface morphology of MicroPCM, (d) and (f) broken microcapsule, (e) magnified broken shell of the microcapsule, and (g) schematic model of shell [54].

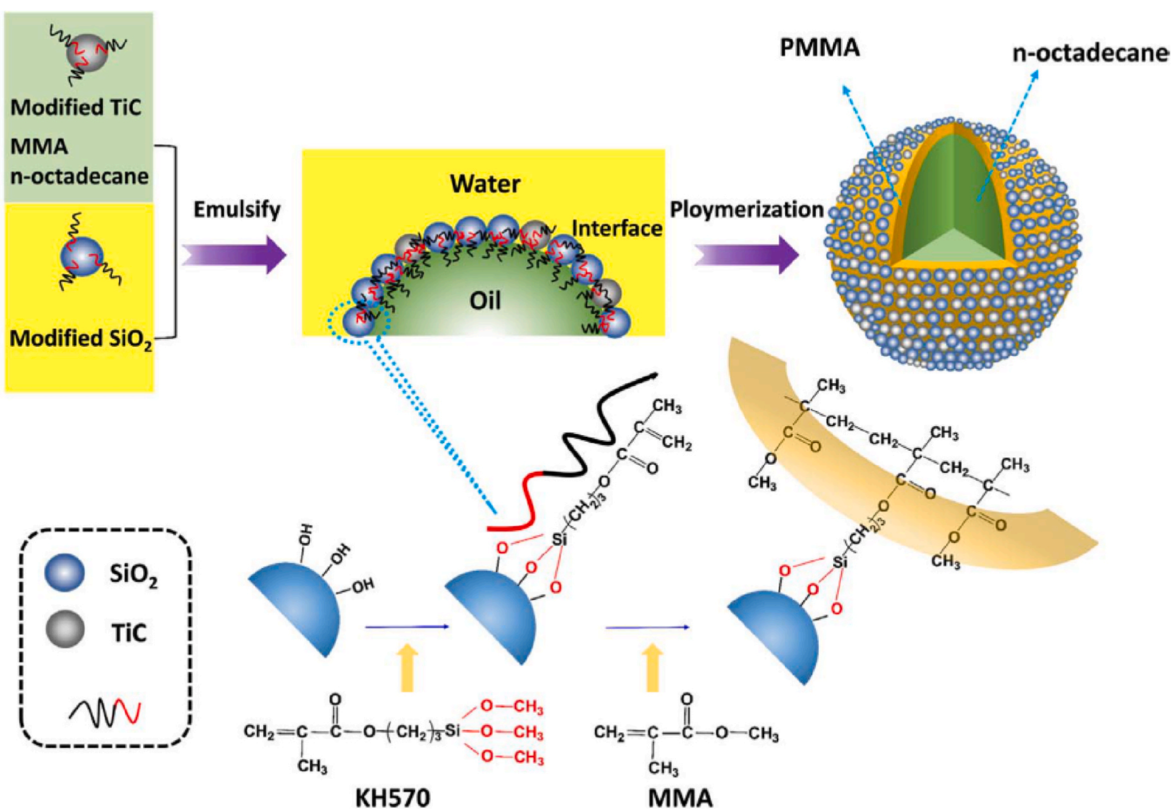


Fig. 8. Schematic of the fabrication process of polymer-SiO_2/TiC hybrid shell microcapsules [57].

PCM consisting of aluminum foam as a support structure and paraffin wax as PCM. The result showed that optimal foam porosity and pore size should be selected for improved thermal performance of composite PCM. Paraffin/metal foam composite PCMs using nickel and copper foam were prepared by Xiao et al. [79] through a vacuum assistance process. The result showed a significant enhancement in thermal conductivity and a slight shift in phase change temperatures. But the overall heat transfer was reduced by 14–24% and 22–30% for nickel foam and copper foam composites, respectively. The same authors reported that the effective thermal conductivity was improved by the increase in the porosity of the metal foam [80]. Atal et al. [81] experimentally/numerically studied the porosity effect on the PCM composite and reported that fewer porosity results in expedited charging/discharging of PCM. Fleming et al. [82] performed an experimental and

Table 1
Summary of the microencapsulated SSPCM from the literature.

PCM	Shell	Preparation method	Loading (%)	Melting point (°C)	Latent heat (Jg^{-1})	Ref.
Caprylic acid	Urea + melamine	Simple Coacervation	59	–	93.9	[59]
n-octadecane	Silica	Sol-gel	–	–	–	[60]
Stearic acid	Silica	Sol-gel	90.7	53.5	171	[61]
n-octadecane	Silica	Interfacial polycondensation	–	–	–	[62]
Paraffin	Silica	In-situ dehydration and condensation	–	–	94.4	[63]
Lauric acid	Silica	Interfacial polymerization	–	–	186.6	[64]
n-tetradecane	Acrylonitrile-styrene	Phase separation	84.5	6.02	142.3	[65]
n-octadecane	Calcium carbonate	Self-assembly	40.35	29.19	84.37	[66]
n-hexadecane	Polyurea-Melamine-formaldehyde	–	62.36	17.42	147.93	[67]
Butyl stearate	Polyurea	Interfacial polycondensation		29	80	[68]
n-octadecane	Polyurethane	Interfacial polycondensation	40–70	29.8–31	115–117.5	[69]
n-octadecane	Polyurea	Interfacial polycondensation	70	28.16	158.7	[70]
$Na_2HPO_4 \cdot 12H_2O$	Silica	Interfacial polymerization + Solgel	75.3	–	177	[71]
Na_2SO_4	Silica	–	–	880.2	170.6	[72]
Li_2CO_3–Na_2CO_3	Silica	–	61.2	498	220	[73]

theoretical analysis of an aluminum foam PCM thermal storage unit. By including aluminum foam, the overall heat transfer was significantly enhanced. Besides, the result showed that higher effective thermal conductivity could be achieved by reducing the pore size. Sundaram and Li [83] investigated the pore size and porosity of PCM-metal foam composites related to heat generation and heat dissipation. By reducing pore size from 100 to 25 μm, the net thermal conductivity was doubled. Through vacuum infiltration, Huang et al. [84] fabricated myristyl alcohol/metal foam composite using nickel and copper foams of different porosity, as shown in Fig. 9. The result showed that higher thermal conductivity enhancement was recorded for smaller pore sizes. Besides, the latent heat was decreased by the addition of metal foams. Qiu et al. experimentally investigated the modified fly ash-based SSPCM and its thermal properties. Lauric acid impregnated into the porous structure of the modified fly ash possessed improved latent heat and thermal conductivity [85].

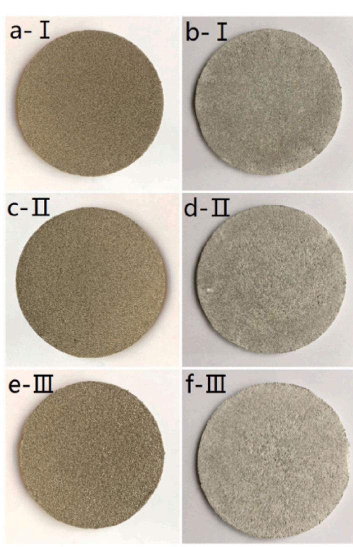
Previous research on metal foams/matrix supports for SSPCM showed that the heat transfer improvement due to the increase in thermal conductivity is significant in metallic support-based SSPCM. Optimizing the pore size and density is essential to develop an efficient metal foam/matrix-supported SSPCM. It is one of the traditional techniques used for SSPCM preparation, but the increase in weight of the TES system is an adverse effect. More research is needed to develop lightweight, efficient metal foam/matrix-supported SSPCMs.

2.3. Carbon-based support structures

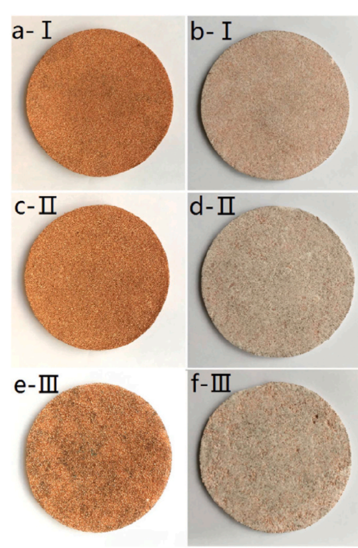
Carbon-based support structures for SSPCMs are highly stable and possess excellent thermal properties such as high thermal conductivity and heat storage ability.

2.3.1. Porous carbon

Porous carbons are materials with high surface area, and the pores are three types, namely macropores, mesopores, and micropores. PCM can be loaded into these pores and can be used as an efficient SSPCM for TES. Due to its stable porosity and good thermal conductivity, better heat transfer can be ensured. Many types of works are available on porous carbon as support structures for SSPCM. Su et al. [86] developed a honeycomb-like porous carbon form PCM with paraffin. The developed form stable PCM exhibits high encapsulation capacity and reliability. With the high loading rate, the latent heat enthalpy was measured as 146.1 Jg^{-1}. A composite PCM with porous carbon sphere and n-octadecane was developed by Ji et al. [87]. The thermal conductivity of the PCM was improved by the addition of porous carbon and showed excellent thermal stability. Shape stabilized porous carbon/-paraffin PCM was prepared by Liu et al. [88] to reduce the room



(a) Nickel foam



(b) Copper foam

Fig. 9. Image of nickel and copper foam before and after preparation with a different pore size (I: 40 PPI, II: 70 PPI, III: 90 PPI) [84].

temperature fluctuations. The porous carbon was prepared by using tin tetrachloride and sodium polyacrylate through the template method. The phase transition rate of the PCM was effectively improved by the porous carbon. The latent heat of fusion was measured as 35 Jg^{-1}. Feng et al. [89] synthesized a novel SSPCM using hierarchical porous carbon with 90 wt% loading capacity. PEG and stearic acid were used as PCM to load in the porous carbon. The addition of porous carbon considerably reduced the supercooling of the PCMs, and heat storage and release were effectively performed. Zhang et al. [90] reported a 3D porous carbon foam-based SSPCM for photothermal conversion. Graphene oxide was used as a surface modifier and paraffin and PEG10000 as PCM. The thermal conductivity of paraffin composite PCM was improved by 300%, with a latent heat of 111.53 Jg^{-1} and photothermal conversion efficiency of 86.68%. SSPCM of PEG encapsulated in porous carbon was developed by Zhao et al. [91]. They reported improved thermal conductivity of the SSPCM. Through vacuum impregnation, 85.36 wt% of PEG absorption was achieved without leakage and presented good thermal stability after 200 thermal cycles. The schematic of porous carbon and SSPCM preparation are shown in Fig. 10.

2.3.2. Expanded graphite (EG)

Expanded graphite is a layered structure of graphite with interlayer space. PCM can be filled in these interlayer spaces and pores, thereby acting as a good support structure for SSPCN. The research works on expanded graphite-based SSPCM are discussed in this section. Rathore et al. [92] investigated the thermophysical properties of an SSPCM with different wt% of expanded graphite and expanded vermiculated as a support structure and a low-cost commercial PCM OM37. SSPCM with 7 wt% expanded graphite melts 25.9% and solidifies 19.2% faster than the pure PCM. The thermal conductivity was improved by 114.4% and showed good stability even after 1000 thermal cycles. Rakkappan et al. [93] developed a composite PCM of 1-decanol with expanded graphite and studied the thermal performance of the air-conditioning application. The prepared composite PCM consisted of 84.99% 1-decanol, and the thermal conductivity was improved by 16.33 times. Using the composite PCM reduced the chiller operating time by 81.85%. Modified expanded graphite-based composite PCM with hydrated salt ($MgCl_2 \cdot 6H_2O$) as PCM was prepared by Zhou et al. [94]. There was no change in melting point, and the supercooling was reduced from 11.2 °C to 1.3 °C. The test result of the thermal reliability of the composite was good.

A high-performance SSPCM with expanded graphite and paraffin PCM was synthesized by Jin et al. [95]. The addition of expanded graphite significantly improved the ability of heat storage and release. After 60 thermal cycles, it showed a 2% reduction in latent heat of fusion. A composite PCM with a ternary eutectic PCM (lauric acid-myristic acid-stearic acid) and expanded graphite were developed by Liu et al. [96] for building application. The melting and freezing point was 29.05 °C and 29.38 °C, and corresponding latent heats were 137.1 Jg^{-1} and 131.3 Jg^{-1}, respectively. The composite PCM was thermally stable even after 1000 thermal cycles. Palmitic acid/expanded graphite form stable PCM with 80:20 wt/wt% was prepared by Sari et al. [97]. The melting and solidifying temperatures were measured as 60.88 °C and 60.81 °C, respectively. The latent heat of melting and freezing was estimated as 148.36 Jg^{-1} and 149.66 Jg^{-1}, respectively. The thermal conductivity of the composite PCM was 2.5 times higher than the pure PCM and showed good thermal reliability up to 3000 thermal cycles. For low-temperature applications, the paraffin/expanded graphite composite was experimentally studied by Lu et al. [83]. The thermal conductivity of composite PCM with 5 wt% expanded graphite was measured to be 1.492 $Wm^{-1}K^{-1}$. The melting temperature was improved, and the addition of expanded graphite decreased the freezing temperature. Zhou et al. [98] developed a composite PCM with xylitol/expanded graphite, and the thermal properties were studied. The results showed that the thermal conductivity was improved 9.29 times, and the latent heat of melting and freezing was 209.73 Jg^{-1} and 192.18 Jg^{-1}, respectively. The expanded graphite and composite PCM preparation process was presented in Fig. 11.

A composite material consisting of sodium acetate trihydrate as PCM with expanded graphite supporting material was synthesized by Fu et al. [100]. In this composite, glycine was added as a temperature regulator. The phase change temperature of the composite PCM was 47.14 °C with a latent heat value of 214.7 Jg^{-1}. The measured thermal conductivity and degree of supercooling were 6.4 $Wm^{-1}K^{-1}$ and 1.49 °C, respectively. Fang et al. [101] prepared a form-stable PCM with $Na_2HPO_4 \cdot 12H_2O$/expanded graphite for radiant floor heating. The modified expanded graphite prevents the PCM from leaking during the phase change process. A high phase change enthalpy of 158.5 Jg^{-1} and a low degree of supercooling of 2.29 °C was evidenced for the prepared form stable PCM. Also, it shows good thermal reliability with thermal conductivity of 1.007 $Wm^{-1}K^{-1}$. Zou et al. [102] developed an SSPCM with $CaCl_2 \cdot 6H_2O$ and expanded graphite of different mesh sizes (50, 80, 100).

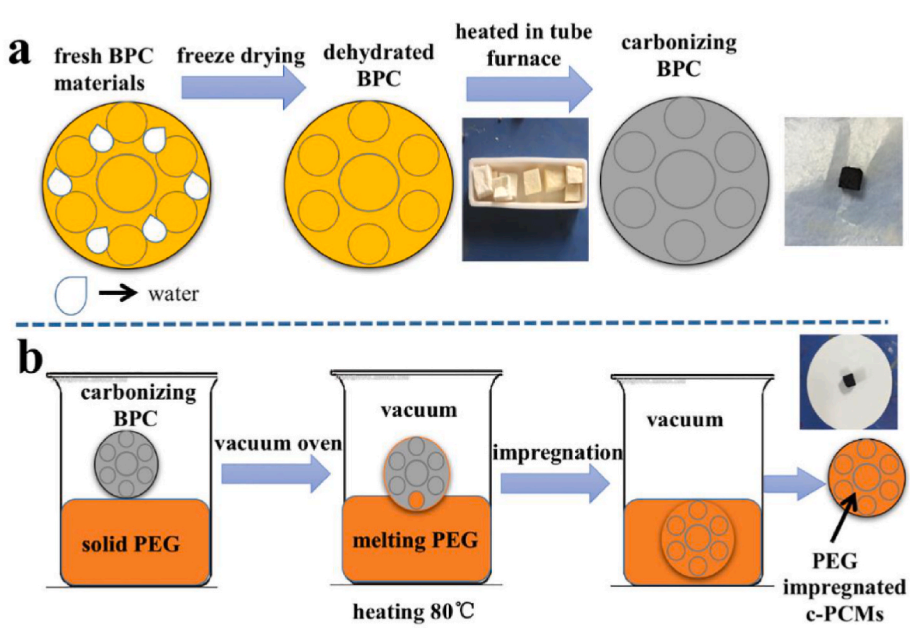


Fig. 10. (a) Schematic of porous carbon synthesis and (b) SSPCM synthesis.

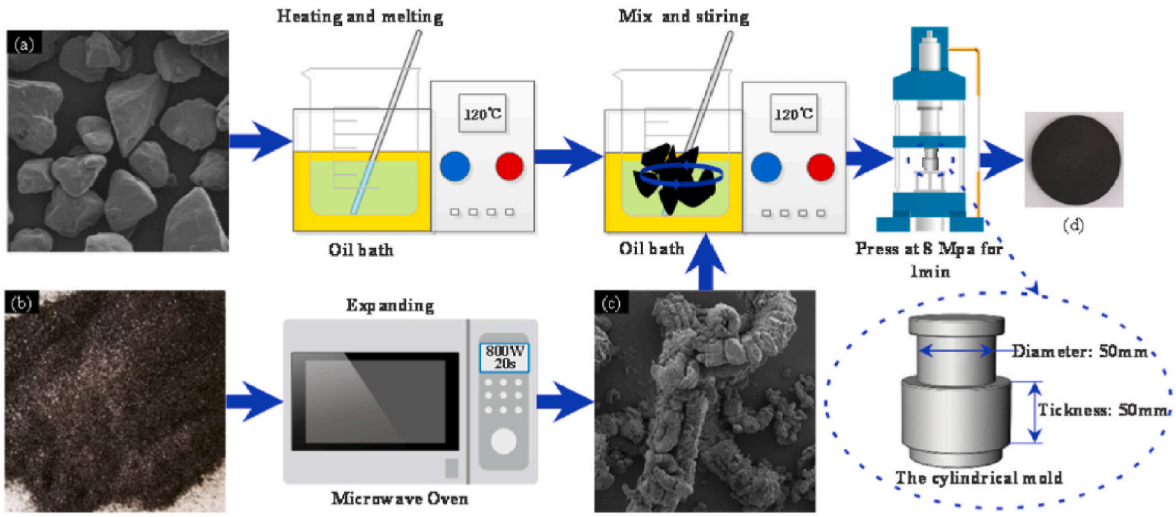


Fig. 11. Fabrication procedure of xylitol/EG composite PCMs: (a) SEM micrograph of pure xylitol, (b) raw expandable graphite, (c) SEM of EG, (d) composite PCM sample [99].

The adsorptive rate of expanded graphite with 50 mesh size was higher than the other and possessed higher thermal conductivity. The addition of expanded graphite reduced the supercooling behavior in the PCM. A composite PCM of high-temperature eutectic chloride salt ($NaCl–KCl–MgCl_2$) using α-alumina/expanded graphite was prepared by Ran et al. [103]. The results showed that the PCM was uniformly distributed in the pores of the support structure with better chemical compatibility. The latent heat was decreased with the increase in alumina and expanded graphite concentration. The thermal conductivity was improved more than 6 times and exhibited good thermal properties even after 300 thermal cycles. Ning et al. [104] prepared an SSPCM by adsorbing a eutectic salt ($K_2HPO_4·3H_2O–NaH_2PO_4·2H_2O–Na_2S_2O_3·5H_2O–H_2O$) in expanded graphite through an impregnation process with an adsorption capacity of 80.71 wt%. The phase change temperature, latent heat, and degree of supercooling were −5.3 °C, 161.8 Jg^{-1}, and 1.83 °C, respectively. The thermal conductivity was improved 13.3 times with improved thermal reliability after thermal cycling. A composite PCM with a medium-temperature ternary nitrate PCM ($KNO_3–LiNO_3–Ca(NO_3)_2$) and expanded graphite was investigated by Lu et al. [105]. The thermal conductivity was improved, and the decomposition temperature was reduced by the addition of expanded graphite. The corrosion compatibility with six metals was tested, and stainless steel 304 and 316 L were found suitable for storage containers. PEG/expanded graphite form stable PCM was prepared by Yang et al. [106] through the vacuum impregnation process using mesoporous EG. The phase change occurs in the temperature range of 18.89–25.93 °C with a latent heat of 97.56–98.59 Jg^{-1}. The thermal stability of the prepared form stable PCM was ensured through TGA.

Graphene oxide-coated tetradecanol/expanded graphite was prepared by Chi et al. [107]. In this work, both microencapsulation and porous carrier techniques were implemented together. The melting and freezing temperatures were 35.2 °C and 34.9 °C, with latent heat of 189.5 Jg^{-1} and 187.9 Jg^{-1}, respectively. Song et al. [108] performed an experimental and numerical investigation on dodecane/expanded graphite SSPCM. The results showed 15 times improved thermal conductivity than pure PCM. Besides, the numerical results were in good agreement with the experimental results. Similarly, another experimental and numerical study on expanded graphite/hydrated salt PCM was performed by Wu et al. [109]. The proposed numerical model helps to predict the thermal conductivity of the carbon matrix composite more precisely. Also, the composite PCM with expanded graphite showed better heat transfer properties at an improved rate. A composite PCM with low melting point alloy/expanded graphite with a mass ratio of 9:1 was developed by Liu et al. [110] for a solar water storage system. Zhao et al. [111] prepared an ethylene glycol/paraffin composite PCM, and the EG size effect on the composite PCM structure was investigated. The result showed a significantly improved thermal conductivity of about 1360–1695% for large EG particles. While using smaller EG particles, the thermal conductivity improvement was 90%–340%. The process of composite PCM preparation is presented in Fig. 12.

The eutectic salt mixture of calcium chloride hexahydrate-magnesium chloride hexahydrate was adsorbed in an expanded graphite matrix, and the thermal properties were studied by Zhang et al. [112]. The results showed that the phase change temperature, degree of supercooling and thermal conductivity were 27.1 °C, 0.11 °C, and 171.4 Jg^{-1}, respectively. The improved thermal conductivity was measured to be 3.359 $Wm^{-1}K^{-1}$. Yuan et al. [113] studied the supercooling and crystallization behavior of erythritol/expanded graphite composite PCM. In this study, nano-Al_2O_3 was used as a supercooling reducing agent.

2.3.3. Graphene and graphene oxide

Graphene is a thin 2D structure made of carbon atoms and has excellent thermal and electrical conductivity with a wide range of applications. Graphene oxide is the oxidized form of graphene, which can be prepared easily compared to graphene. Due to their significant thermal properties, they are being used as potential support structures for SSPCM. Previous research on graphene and graphene oxide-based SSPCM is discussed in this section. Yan et al. [114] studied the heat transfer performance of erythritol/graphene composite PCM. The thermal conductivity was improved two times that of pure PCM, and the supercooling decreased from 68 °C to 21 °C for the composite PCM with 8 wt% of graphene. Reduced graphene oxide with cobalt oxide nanoparticles was used as a support structure for PEG as PCM and studied its performance for solar to thermal energy conversion by Li et al. [115]. The thermal conductivity improved from about 51.79% to 152.46% compared to pure PCM. The latent heat of the SSPCM was 131.94 Jg^{-1} and exhibited good shape stability with excellent solar thermal energy conversion. Zhang et al. [116] designed stearic acid/graphene oxide-attapulgite aerogel SSPCM with improved thermophysical properties through the hydrothermal process. The developed SSPCM exhibited no leakage and a significant thermal storage capacity of 190.9 Jg^{-1}. Form stable PCM with PEG/reduced graphene oxide was prepared by Ding et al. [117] through melting infiltration and hydrothermal reduction processes, as shown in Fig. 13. The PCM mass content in SSPCM prepared through hydrothermal reduction was 85.6% with a

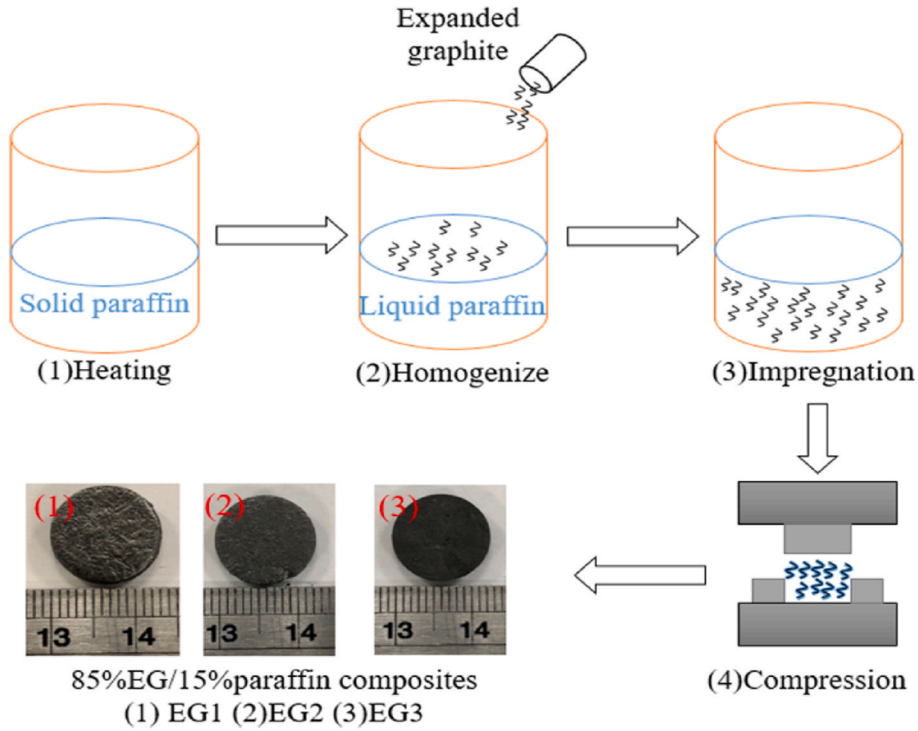


Fig. 12. Composite PCM sample preparation [111].

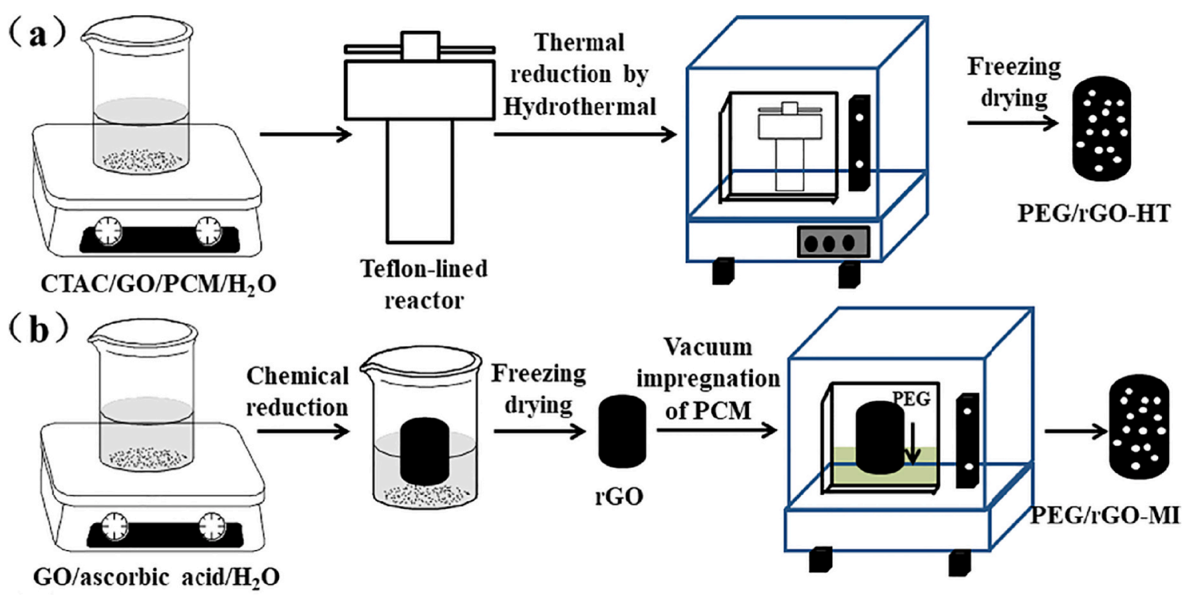


Fig. 13. Schematic of the preparation of PEG/reduced graphene oxide SSPCM through (a) Hydrothermal reduction (b) Melting infiltration [117].

latent heat of 139.4 Jg^{-1}, whereas the SSPCM prepared through melting infiltration was 96.6% with a latent heat of 205.2 Jg^{-1}.

Li et al. [118] successfully developed lauric acid encapsulated in reduced graphene oxide and n-doped porous carbon supporting scaffold. The prepared SSPCM showed an excellent photothermal response. Jin et al. [119] prepared a composite PCM with capric acid-stearic acid eutectic PCM, montmorillonite, and graphene for building TES. The composite PCM was thermally stable up to 300 thermal cycles and showed a 156.4% increase in thermal conductivity for 2 wt% of graphene content. Carboxyl functionalized graphene/paraffin wax composite PCM was prepared by Pramod et al. [120] and studied the effect of concentration. As the thermal conductivity improved, the charging time was reduced up to 20%–30%. A composite PCM consisting of graphene oxide, carbon nanotubes, and paraffin wax was prepared by Chen et al. [121] for solar TES applications. The thermal conductivity of the composite PCM was measured to be 0.46 $Wm^{-1}K^{-1}$ with a solar absorptance of 0.952. N-eicosane/multilayer graphene composite PCM was developed by Zhang et al. [122] for electro-thermal conversion and storage. The inclusion of multilayer graphene enhanced the thermal conductivity and 45 wt% of graphene inclusion showed excellent electro-thermal conversion performance. Joseph and Sajith [123] prepared a graphene/paraffin nanocomposite for electronics thermal management. With the addition of 0.5 wt% of graphene, the thermal conductivity was enhanced up to 60%. PEG encapsulated in graphene aerogel was successfully developed by Liao et al. [124]. The prepared composite PCM possesses high thermal conductivity and high latent heat. After thermal cycles, the composite PCM shows excellent stability. The SSPCM of PEG with graphene foam and carbon nanotubes was prepared by Yu et al. [125]. The thermal conductivity was improved by 118% with a melting latent heat of 128.7 Jg^{-1}. Li et al. [126] prepared graphene-based composite PCM through one-step synthesis. The loading of PEG was about 95 wt%, and no leakage was evidenced. The latent heat of the composite PCM was 163.5 Jg^{-1}, with a solar conversion efficiency of 93.7%. Ge et al. [127] investigated the thermal properties of shape stabilized epoxidized methoxy PEG/functionalized graphene oxide composite PCM. A superior stabilization was exhibited for the composite PCM after 100 thermal cycles without leakage up to 105 °C. Ashiq Ali et al. [128] investigated the heat transfer enhancement of paraffin wax using nanographene for industrial helmets. The NG/paraffin composite showed a 146% improvement in thermal conductivity with a minor reduction in latent heat.

2.3.4. Carbon nano-tubes (CNT) and multi-walled CNT (MWCNT)

CNTs or SWCNTs are nanoscale hollow tubular structures made of carbon atoms. MWCNTs refer to the nested structure of SWCNTs. Both CNTs and MWCNTs possess exceptional thermal properties and are being investigated as a support structure for SSPCM. The previous works on CNT and MWCNT-based SSPCMs are introduced in this section. Cheng et al. [129] prepared CNT microcapsule PCM for the application of thermal comfort in buildings with a melting point and latent heat capacity of 28.97 °C and 112.25 Jg^{-1}, respectively. The composite PCM performed well in reducing indoor temperature fluctuations. Kuziel et al. [130] developed ultra-long CNT-paraffin composite PCM with superior thermal properties. The thermal conductivity of the composite PCM was improved by 161% compared with the pure PCM and the phase change enthalpy increased by 6.3%. The thermophysical properties of paraffin-CNT composite PCM were studied by Hong et al. [131]. With the addition of 3% CNT, the thermal conductivity in solid and liquid phases increased by 30.3% and 28.5%, respectively. Also, the melting and freezing latent heat were reduced by 8.9% and 9.3%, respectively. Zhu et al. [132] investigated the effect of geometry on the thermal storage performance of CNT/paraffin SSPCM. The results revealed that the thermal storage capacity improved by 19.6% more than the theoretically estimated value. In the composite PCM, 93 wt% of PCM was loaded and showed no leakage. The melting latent heat was 218.56 $kJkg^{-1}$, and about 99% was retained for 500 thermal cycles. Multifunctional paraffin wax/CNT sponge composites with a high latent heat of 248.8 Jg^{-1} were developed by Lu et al. [133]. The prepared composite PCM can be used for thermal management and electromagnetic shielding of electronic devices. Zhang et al. [134] prepared composite PCM with palmitic acid-stearic acid eutectic mixture and CNT at different concentrations ranging from 1% to 8%. The results showed that the thermal release rate was increased, and the addition of CNT decreased the storage rate. The photographs of the prepared composite PCMs are shown in Fig. 14.

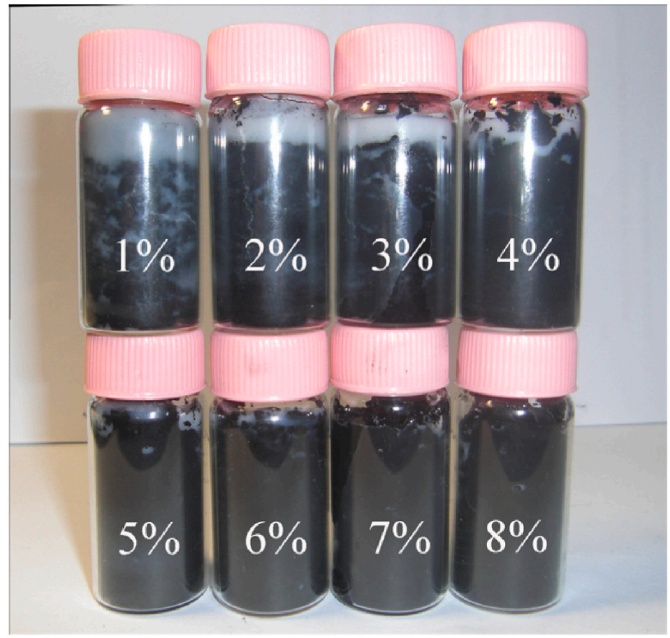


Fig. 14. Photographs of the prepared PA–SA/CNTs composite PCMs [134].

A novel beeswax/MWCNT SSPCM was prepared by N. Putra et al. [135] through the impregnation process, as shown in Fig. 15. They investigated the effect of using pristine, ball-milled, and acid-treated CNT for the preparation. The pretreatment of CNT influenced the form-stable PCM. The thermal conductivity was improved, and no change in phase change temperature was evidenced.

Summarized carbon-based SSPCMs from the literature are presented in Table 2. Organic PCMs are widely studied with carbon-based support structures as their thermal conductivity needs to be improved. Paraffin is majorly investigated for SSPCMs due to their wide range of phase change temperature availability and high thermal storage capacity. High thermal conductivities were reported for the SSPCMs using EG and graphene support structures. But the economic viability of using graphene is less and technological advancements are to be explored further. EG shows significant improvement as well as high PCM loading and thermal performance. The thermal conductivity reported for SSPCMs with CNT support structure was less compared to SSPCMs with EG and graphene support structures.

The various carbon-based supports for SSPCM are discussed in this section. The carbon-based supports such as porous carbon, EG, graphene/graphene oxide, and CNT/MWCNT are employed in the preparation of composite PCMs due to their advanced heat transfer and energy conversion properties and are extensively studied by the researchers. The TES capacity of the carbon-based SSPCM depends on the size and structure of the carbon support. EG is a promising candidate among the different carbon supports due to its high TES capacity. On the other hand, CNTs and graphene show better performance in terms of heat transfer. High-performance carbon-based SSPCM can increase the use of TES technologies and promote renewable and sustainable energy utilization. Despite the recent advancements, more in-depth research on carbon-based SSPCM is necessary to address the critical issues such as heat transfer and energy conversion that will result in the advanced SSPCM for TES.

2.4. Polymer matrix

Microencapsulation of PCM is a complicated process and expensive one. Selecting the right material for shell material will reduce the preparation cost. Therefore, the polymer matrix is used for the low-cost confinement of PCMs. The different polymers used as support structures for PCMs are polyacrylates, polyolefins, polysaccharides, styrene block copolymers, and polyurethane. Hu [168] presented the recent advancements in polymeric PCM composites and their applications. Pandey et al. [169] encapsulate n-eicosane in an amphiphilic polymeric matrix through a double emulsion technique. The encapsulation efficiency was above 95% and latent heat was about 160 Jg^{-1}. The hydrophilic nature of the shell material provides good wettability and efficient thermal management, having the potential in space conditioning applications. Li et al. [170] developed a comb-like polymer (poly (tetradecyl acrylate)) blended with different percentages of boron nitride. The SSPCM with 20% boron nitride showed a 113.8% enhancement in thermal conductivity. Besides, the thermal storage and released rate improvement were about 208% and 110%, respectively. The thermal performance of SSPCM-coated fabric was also evaluated. Barreneche et al. [171] developed and studied an SSPCM that consists of the

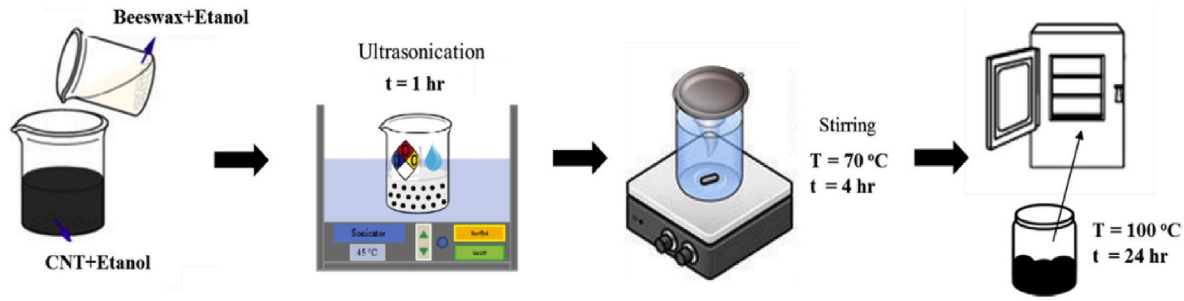


Fig. 15. Composites preparation by impregnation process [136].

Table 2
Summary of the carbon-based SSPCM from literature.

PCM	Carbon support	Melting point (°C)	Latent heat (Jg^{-1})	Thermal conductivity ($Wm^{-1}K^{-1}$)	Ref.
Paraffin	EG	48.9	161.5	–	[137]
Paraffin	EG	40.2	178.3	0.82	[138]
Paraffin	EG	61.3	117.5	3.83	[139]
Paraffin	EG	52.2	170.3		[140]
Paraffin	EG/epoxy	–	–	–	[141]
Stearic acid	EG	53.1	155.5	–	[142]
Sebacic acid	EG	128	187	5.35	[143]
Palmitic acid	EG	60.9	148.4	0.60	[144]
$CaCl_2{\cdot}6H_2O$	EG	27.1	118.7	3.20	[145]
KNO_3–$NaNO_3$	EG	89	104.7	50.8	[146]
$Ca(NO_3)_2$–$NaNO_3$	EG	216.8/280.8	89.79	5.66	[147]
Stearic acid/Benzamide	EG	65.63	176.03	4.177	[148]
PEG6000	Graphene	63.1	165.8	–	[149]
Palmitic acid	Graphene	68.4	126.3	2.26	[150]
Erythritol	Graphene	120.01	338.6	1.122	[151]
Stearic acid	Graphene	51.5	51.3	1.05	[152]
Stearyl alcohol	Graphene	–	210	0.91	[153]
Paraffin	Graphene	60.8	200	0.62	[154]
Palmitic acid	Graphene oxide	59.7	100.2	1.24	[155]
PEG8000	Graphene oxide	65.3	158.2	–	[156]
PEG	Graphene oxide/boron nitride	65.5	157.7	1.72	[157]
PEG	Graphene oxide/boron nitride	65.7	145.9	1.84	[158]
Paraffin	CNT	28	47.7	0.30	[159]
Stearic acid	CNT	29	111.8	–	[160]
PEG10000	CNT	51.6	100.5	0.33	[161]
PEG6000	CNT	53.3	135.1	0.46	[162]
Paraffin wax	CNT	52	163.8	0.32	[163]
Paraffin	SWCNT	37.88	–	2.7	[164]
PEG	MWCNT	61.1	92.6	–	[165]
PEG	MWCNT	55.6	151.9	0.429	[166]
Erythritol	MWCNT	–	–	0.9779	[167]

polymer matrix (polyethylene-co-vinyl acetate copolymer), PCM (RT21), electrical arc furnace dust (EAFD), and zinc stearate. The phase change temperature and latent heat of the SSPCM were 20.21 °C and 10 $kJkg^{-1}$, respectively. The developed material can be used in building construction as acoustic insulation. Zou et al. [172] used a super absorbent polymer as a support structure for a eutectic hydrated salt ($Na_2HPO_4{\cdot}12H_2O$–$K_2HPO_4{\cdot}3H_2O$) PCM while preparing the SSPCM. $Na_2SiO_3.9H_2O$ was used as a nucleating agent. The eutectic hydrated salt

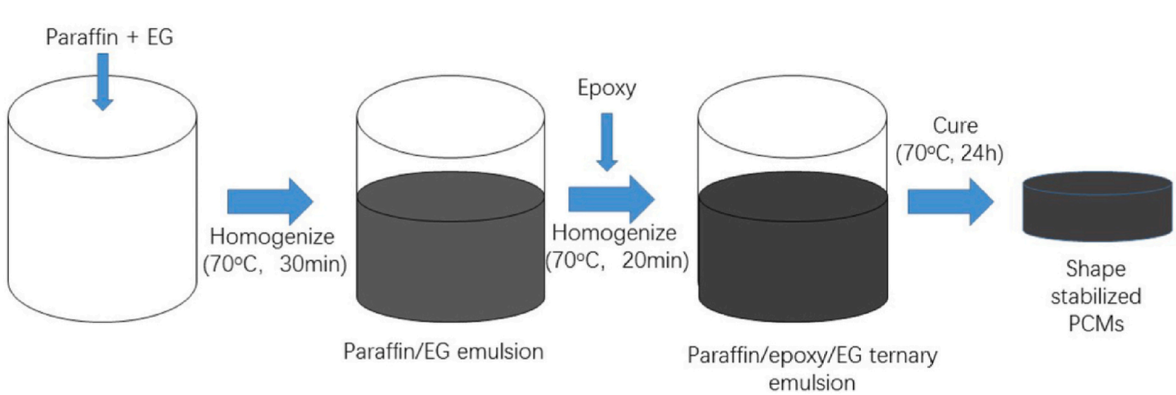


Fig. 16. Process of preparing SSPCM [141].

PCM was loaded in the polymer support with a mass fraction of 12% through physical interaction. The melting point and latent heat were measured to be 24.13 °C and 172.7 Jg^{-1}, respectively. The prepared SSPCM was stable and thermally reliable up to 100 thermal cycles. Wang et al. [141] developed a novel SSPCM with an extended life cycle and compact structure using epoxy matrix and paraffin. The SSPCM was prepared with a 1:1 mass ratio of epoxy and PCM and showed good mechanical and thermal stability without leakage. The preparation process of SSPCM is illustrated in Fig. 16.

The thermal storage and release ability of an SSPCM consisting of styrene-b-(ethylene-co-butylene) as a supporting network and hexadecane as PCM coated with low-density polyethylene was studied by Chriaa et al. [173]. The composite PCMs were prepared with a different mass fraction of hexadecane, and the highest latent heat of 162.37 $kJkg^{-1}$ was measured for composite PCM with 80% hexadecane and showed good form stability. An organic polymer SSPCM with porous triamide-linked polymer and PEG2000 was developed by Andriamitantsoa et al. [174] with a maximum PCM loading capacity of 85%. The melting and freezing latent heat were 155 $kJkg^{-1}$ and 141.7 $kJkg^{-1}$, with a melting and freezing point of 53.13 °C and 29.67 °C, respectively. The SSPCM was found to be stable over 50 thermal cycles. Yang et al. [16] fabricated a flexible SSPCM using cross-linked polymer through swelling technique and used for thermal management. The SSPCM possesses excellent mechanical properties. The composite PCM film can reduce the electronic device and building temperature by 5.5 °C and 9.3 °C, respectively. Li et al. [175] prepared porphyrin-ferrocenyl conjugated microporous polymer nanospheres through solvothermal synthesis and used it as a support structure to hold 1-octadecanol. The result showed a high latent heat of 153.8 Jg^{-1} with good shape stability and durability. The thermophysical properties of epoxy resin composite SSPCM were analyzed by Ma et al. [176]. The SSPCM with Pentadecane PCM possesses good thermal storage performance, stability, and mechanical properties. But, the measured thermal conductivity decreased with an increase in temperature. Shen et al. [177] developed a composite PCM of polyethylene glycol and a double network support material consisting of phosphorylated polyvinyl alcohol/graphene aerogel. The result showed that the peak heat release rate of composite PCM decreased compared to pure PCM.

Summarized SSPCMs with polymer matrix support from the literature are presented in Table 3. It can be seen from the literature that polymer matrix support structures are majorly studied with organic PCMs and few inorganic hydrated salts. The results showed that the SSPCMs with polymer matrix support structures exhibited good thermal storage capacity. But the thermal conductivities of the polymer matrix-based SSPCMs are not reported for all the developed materials. Since it is flexible and more stable, it can be used for different applications in various fields.

The previous research on polymer matrix-based SSPCM elucidated that the polymer supports are lightweight and highly stable. In particular, due to its flexibility, it can be used for a wide range of applications. The polymer matrix is cheaper than other techniques and exhibits good stability. However, more research is also needed to improve thermal conductivity and develop a highly conductive polymer matrix to develop an efficient SSPCM.

2.5. Metal organic framework

Metal organic frameworks (MOFs) are hybrid porous materials that consist of a regular array of metal ions surrounded by organic linker molecules. MOFs have excellent chemical and structural stability, ultrahigh surface area, and flexibility. Due to these superior properties, MOFs can be used as a potential support structure for SSPCM. Many previous research works discuss the MOF-based SSPCM. Chen et al. [190] reviewed MOF-based PCM for TES applications, providing in-depth information on MOF structure and its performance. Zahir et al. [191] developed a hybrid polyMOF material using PEG/UiO-66 with tunable characteristics and unique morphology. The prepared SSPCM shows enhanced thermal conductivity more than the pristine PCM. The impact of novel MOF-based composite PCM on building energy consumption was investigated by Qin et al. [192]. The result showed that the energy-saving potential of using MOF-PCM was up to 35.2%. Pornea et al. [193] synthesized a hybrid dual-MOF encapsulated PCM for efficient photothermal conversion and storage. The improved photothermal conversion efficiency was about 96.2%. In addition, Hu et al. [194] developed a porous EG@Ni-MOF via the hydrothermal method, and it was further used to prepare composite PCM through physical blending. Fig. 17. Shows the preparation process of (a) Ni-MOF and (b) EG@Ni-MOF. The result showed a 3.21 times improvement in thermal conductivity with a latent heat of 166.6 Jg^{-1}. The SEM images of EG, Ni-MOF, and composite PCMs are presented in Fig. 18.

Li et al. [195] investigated the effect of MOF-derived hierarchical Co_3O_4/EG on stearic acid as PCM. The effective heat transfer and enhanced thermal properties were due to the high surface area and abundant pore volume of the Co_3O_4/EG. Kuai et al. [196] studied the thermal characteristics of the multilayered structural MOF-EG/OC composite PCM and reported a high latent heat of 187.04 $kJkg^{-1}$. Previous studies on MOF-based SSPCM are advantageous in their chemical and structural versatility for many interdisciplinary applications. However, despite the advantages, there are several challenges while using pristine MOF structures, such as low loading content and low latent heat. This is due to the small pore structure of MOF support which results in a strong nanoconfinement effect for PCM loading. More research on pore modification techniques should be investigated to explore this topic further.

3. Applications of SSPCM

It is well known that PCMs have a wide range of applications such as solar thermal, building thermal comfort, heat recovery, power generation, thermal management in electronics, healthcare, textiles, etc.

Table 3
Summary of the SSPCMs with polymer matrix support from the literature.

PCM	Support	Melting point (°C)	Latent heat (Jg^{-1})	Thermal conductivity ($Wm^{-1}K^{-1}$)	Ref.
Paraffin	Poly (styrene-butadiene-styrene)	–	81.63–165.2	–	[178]
Hexadecanol	Polyurethane	39.6, 50.3	229.5	–	[179]
Cetyl alcohol	HDPE	50	149.02–212.42	0.47 (solid) 0.33 (liquid)	[180]
1-octadecanol	Hierarchical porous polymer	58.5	169.1	–	[181]
PEG	Imine-linked micron-network polymer	–	164.9	–	[182]
n-carboxylic acids	Conjugated microporous polymer	–	103.3–171	–	[183]
Palmitic acid	Polypyrrole	–	166.3	–	[184]
Myristic acid	Polyaniline	–	150.63	–	[185]
Paraffin	Epoxy	36, 60	152.6	–	[186]
PEG	Triallyl isocyanurate	31.16–57.14	110–137	–	[187]
Paraffin	Styrene-ethylene-propylene-styrene	50.2	178.8	–	[188]
Paraffin	Olefin block copolymer	–	145.26	1.564	[189]

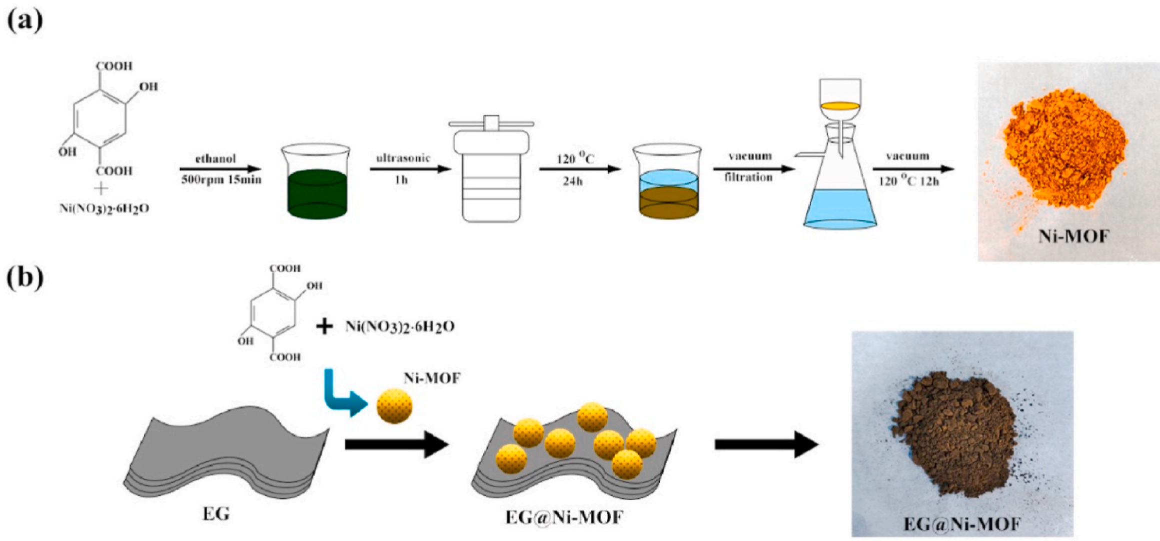


Fig. 17. The preparation process of (a) Ni-MOF and (b) EG@Ni-MOF [194].

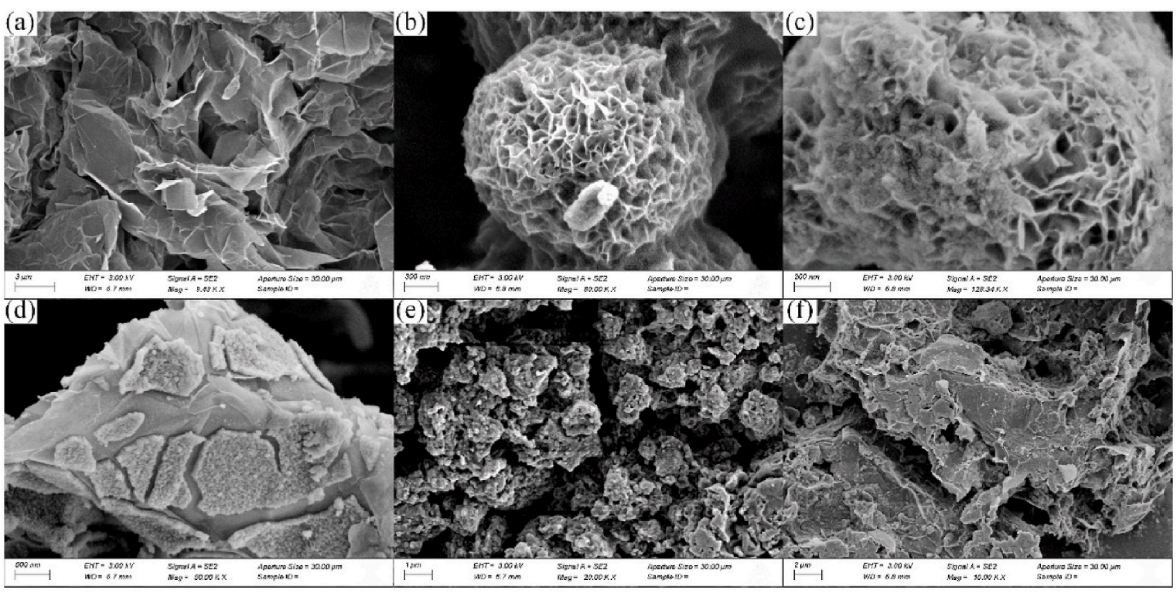


Fig. 18. SEM images of (a) EG, (b) and (c) Ni-MOF, (d) EG@Ni-MOF, (e) SA/Ni-MOF and (f) SA/EG@Ni-MOF [194].

SSPCMs with different support structures are used in different applications due to their excellent heat storage/release capacity and improved thermal properties.

3.1. Building application

Using SSPCM in buildings has considerable energy storage potential and attracted researchers. Zhang et al. [197] evaluated the performance of SSPCM for PCM floor or wallboard and electric underfloor space heating system. Some numerical models were developed and validated for SSPCM applications. Yang et al. [198] investigated the ceramist-based SSPCM for building energy conservation. The concrete-based CPCM for reducing indoor temperature fluctuation. This helps to reduce the heating and cooling energy consumption in buildings. Kim et al. [199] evaluated the SSPCM application for indoor heating load reduction using three identical huts with SSPCM sheets, as shown in Fig. 19. The results showed that optimum installed area and capacity improve the efficiency based on the climate, and it can reduce the heating and cooling energy requirement. In another study by the same author, the effect of heat load reduction by incorporating SSPCM sheets in different Japanese climates was presented [200].

Kong et al. [201] experimentally studied the performance of MEPCM panels in the walls of a building. Eutectic PCM (Capric acid–1-dodecanol) with a melting point of 26.5 °C was used as PCM. The PCM panels reduced the peak temperature and fluctuation in room temperature. In another work [202], the same author numerically investigated the thermal performance of walls and roofs with PCM panels. Numerical solutions for maintaining thermal comfort using PCM panels in walls and roofs were presented. SSPCM layer was introduced in buildings, and the acoustic and thermal performance was evaluated by Barreneche et al. [203]. The SSPCM layer reduces the indoor building temperature by 3 °C and can acoustically insulate up to 4 dB. Besides, the leaching test confirms the non-hazardous nature of the SSPCM sheet. Yao et al. [204] performed a numerical and experimental study on expanded perlite-based SSPCM wallboards in a test cube. The experimental result showed that the SSPCM wallboard reduces the indoor

Fig. 19. Exterior of the test huts [199].

temperature by 2.53 K. An optimal SSPCM wallboard configuration was obtained from the numerical results. Diaconu and Cruceru [205] evaluated a novel composite wall with two different PCM wallboards, which can perform in both hot and cold seasons. The composite wall system reduces the peak heating/cooling loads and provides high energy savings throughout the year. Zhou et al. [206] analyzed a direct gain room with SSPCM plates using an enthalpy model. The results elucidated that the optimal melting temperature and latent heat of the SSPCM should be 20 °C and 90 $kJkg^{-1}$. The SSPCM plates make the structure heavy and increase the nighttime room temperature by 3 °C by storing the solar thermal energy in the daytime. The schematic of the simulated room is shown in Fig. 20, depicting the location of the SSPCM panels in the ceiling, side walls, and floor. In another work [207], the same authors numerically analyzed the SSPCM plates in buildings with night ventilation technique. By storing cold in the nighttime, the daytime maximum temperature was reduced by 2 °C. Besides, the air change per hour should be high compared to the daytime.

Biswas et al. [208] performed field testing on nano-PCM wallboard under different weather conditions. A numerical model was also developed and validated with experimental results. The results showed that the PCM wallboards reduce peak cooling loads and evaluate the energy-saving potential. Ye et al. [209] evaluated the performance of SSPCM in buildings via an energy-saving index under various climatic conditions. The PCM showed better performance in summer than in winter. The expanded polystyrene used as insulation material performs well throughout the year. A PCM embedded floor panel for thermal management of buildings was developed by Royon et al. [210]. To avoid leakage, a polymer-PCM composite was used inside the floor cavities. The numerical and experimental results were compared, and the optimal PCM percentage was identified. Silva et al. [211] developed a window shutter with PCM and performed an experiment in an outdoor test cell, as shown in Fig. 21. In the compartment with PCM, the maximum measured temperature was 37.2 °C which was 16.6 °C lower than the compartment without PCM. The results showed that the PCM helps in thermal regulation in an energy-efficient approach. In another work [212], the same authors studied the performance of window shutters with PCM under summer Mediterranean climate and presented their experimental results.

3.2. Textiles application

SSPCM can be used in textiles for thermal management, but the application is not explored enough. Ghahremanzadeh et al. [213] studied the fastness property improvement of PCM on surface-modified wool fabrics. Cotton and polyester fibers have good adhesion with PEG polymer, but it shows poor adhesion with wool fiber. Therefore, the wool fabric was gaseous fluorinated and chlorine-Hercosett treated and it was examined to have cross-linked PEG with good fastness properties. The result shows that a high surface area is necessary to obtain wool fiber with higher polymer adhesion, resulting in high thermal conductivity, durability, and enhanced felting performance. Ke et al. [214] prepared a porous phase change PEG/polyurethane membrane and evaluated the performance. The different thermophysical properties were investigated and the result showed that the PEG/polyurethane membrane has a porous structure with high phase change enthalpy and suitable phase change temperature. Therefore, the functional polymer

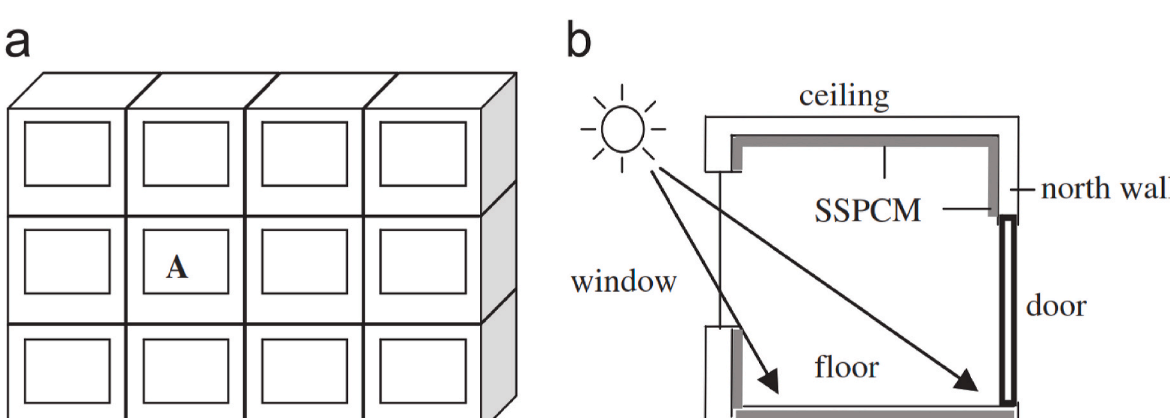


Fig. 20. Schematic of the simulated room (a) location of room A (b) profile of room A with SSPCM [206].

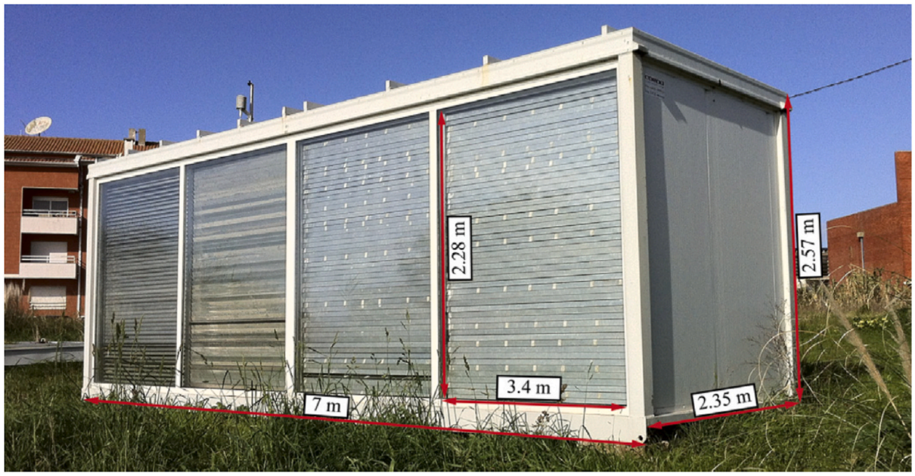


Fig. 21. Window shutters with PCM used in outdoor test cell [211].

can be used as phase change textiles with improved phase change enthalpy.

3.3. Photovoltaic (PV) panel cooling

The performance of the PV panel is significantly affected by the increase in panel temperature. SSPCMs are used to maintain the PV panel temperature at a nominal range so that the PV panel operates at maximum efficiency. Luo et al. [215] performed a numerical and experimental investigation on the thermal management of solar panels using form-stable composite PCM. By using the form stable PCM, the PV panel was maintained below 50 °C for 200 min. The output power was increased by 7.28% during melting. The experimental and numerically simulated temperatures were in good agreement.

The various TES applications of SSPCM are discussed in this section. It can be seen from the previous investigations that SSPCMs have significant potential in thermal management due to their improved heat transfer properties. More specifically, floor heating/cooling in buildings and regulating indoor space temperature are several significant applications. Besides, building materials made with SSPCMs are employed in building integrated thermal management. It can be concluded from this section that the applications of SSPCM have been increasing in recent times, and exploring the possible ways for integrating SSPCMs in various TES applications will significantly contribute to energy conservation and utilize the available energy efficiently.

4. Conclusion and future outlook

Shape stabilization is a promising technique that helps to improve the heat transfer in latent heat TES. This review presents previous research on SSPCM for TES applications. More specifically, the different types of support structures used for the preparation of SSPCM are introduced and discussed, along with various techniques adopted. Furthermore, the effect on thermophysical properties, such as phase transition temperature, thermal conductivity, stability, degree of supercooling, etc., of the SSPCM in the literature is comprehensively discussed and presented. Based on this review work, the following conclusions have arrived.

1. Microencapsulation is one of the best ways of encapsulating the PCM and is used for different applications. Among the different methods, the chemical method is widely used and more specifically polymerization technique is used for microencapsulation. Organic shell materials are flexible, but inorganic shell materials are thermally stable and with good thermal conductivity. Hybrid shell materials possess good thermal stability and are also used in various applications. Encapsulation ratio and core-shell size are essential parameters to be focused on.
2. Metallic supports are stable structures and exhibit good thermal conductivity. Pore size decides the performance of SSPCM, and reduced pore size results in improved thermal conductivity. The optimized pore size and pore density determine the heat storage and retrieval rate of the metal foam-based SSPCM.
3. Carbon-based support structures are versatile, highly stable, and possess excellent thermal properties. It has multifunctional applications due to its advanced properties. Porous carbon is a simple structure into which PCMs are loaded through vacuum impregnation. CNT and MWCNT are nano supports with effective thermal conductivity and high surface area for good heat transport. Graphene and graphene oxide support structures are relatively expensive structures and need advanced mechanisms to develop SSPCM. Carbon-based composite PCMs are highly stable at elevated temperatures and are used in solar thermal applications.
4. The polymer matrix is a low-cost and flexible support used for SSPCM with good mechanical strength. But it has a negative impact on thermal conductivity with an increase in temperature.
5. MOF-based SSPCMs are having excellent stability and high surface area. But low thermal conductivity and poor PCM loading limit its suitability for TES applications which can be rectified by advanced techniques.
6. SSPCMs have huge potential in building thermal management compared with other applications as they have been incorporated into walls, ceilings, and floors of the building structure. Several other applications, such as solar thermal and thermal management of electronic devices, are also explored.

Undoubtedly, SSPCMs are potential candidates for thermal energy storage. However, there are several challenges in preparing and implementing SSPCM in real-time TES applications. Thorough investigations on bulk production and various crucial properties of the SSPCMs have not been executed enough. Therefore, the future work suggestions can be on the following aspects.

- Advancement in the preparation techniques for bulk production and commercialization.
- The stability of the SSPCMs should be studied for long-term applications to avoid performance reduction upon continuous usage.
- Supercooling is reduced by SSPCM preparation but needs further investigation to achieve a higher reduction that makes the suitability of SSPCMs in industrial applications.

- Most research on SSPCM deals with organic PCMs. The suitability of the inorganic PCMs with high latent heat for SSPCM should be investigated to exclude the problems such as phase separation.
- Composite support structures are to be developed with different functional groups for better loading, improved thermal conductivity, and latent heat.
- Researches on the mechanical properties of the SSPCMs are limited. More studies on mechanical properties for different support structures should be executed that provide insight to expand the application area.

This review elucidates that SSPCM has a variety of applications, and it is mainly employed in building structures for thermal management. Existing research works primarily focus on developing new and efficient SSPCMs with different support structures at the laboratory level. Real-time investigation of SSPCM is essential to explore the maximum potential in TES. Besides, it is also reported that the preparation of SSPCM has several adverse effects, such as a reduction in TES capacity and stability. By considering all these factors, future research should focus on developing high-performing SSPCMs with high energy storage capacity and stability and real-time application studies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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